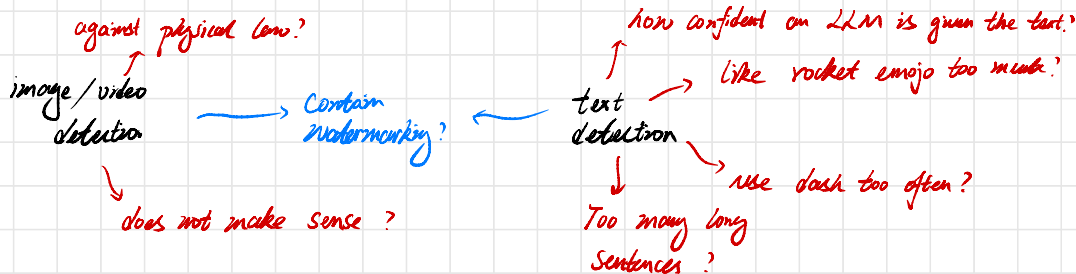


An important question in the age of LLM is to detect if given texts are generated by LLM or written by human.

This exactly the same question is important for image or video generations (although they are arguably easier... becoming harder and harder as well!) Obviously, we can check if the image makes sense, e.g. if the shadows match the objects, if any objects against physical laws etc. another common technique for identifying an image is to add watermarking explicitly or implicitly.



Common text detection method

1. **perplexity**: because LLM predicts the probability for the next word.

if for most tokens in the total AI has high confidence about this token given all tokens before it, then it is likely that the texts are generated by AI.

pros: easy to implement

- Cons:
- 1. most AI companies like OpenAI does not disclose the probability (otherwise their models can be distilled easily)
 - 2. the low confidence could come from model A vs model B not human vs AI
 - 3. most importantly, if the texts are in training dataset, the model will be very confident, e.g. Shakespeare's books...

2. Statistical detection. An LLM often writes differently than human. For example, it could use too many emojis, the sentence lengths are all long (but humans prefer mixing short and long sentences)

example → AI uses dash "-" more often than human.

↓
we can then detect the text by examining the frequency of dash in the texts.

pros: easy to implement

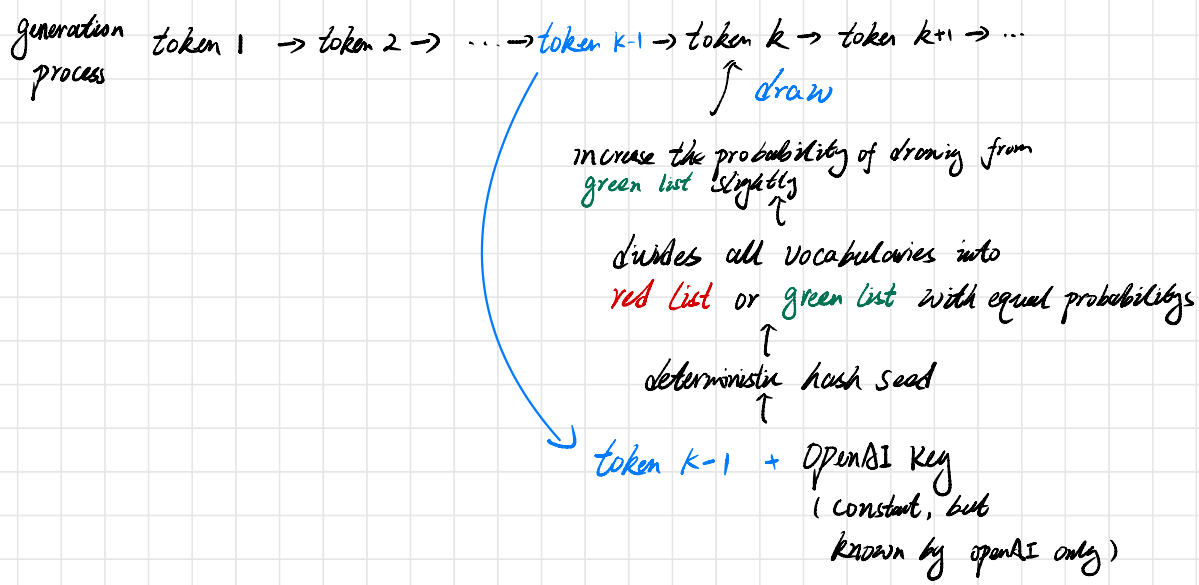
- Cons:
1. not very reliable (could have low detection rate depending on the statistical properties being tested)
 2. people like using dash (or others) are innocent...
 3. Can only detect relatively "easy" patterns.

3. Classifiers. Training one classifier model to decide if the texts are AI generated by feeding it a lot of examples of AI generations and human writings.

pros: easy to use and compared to the statistical method above, it can detect deeper pattern.

- Cons:
1. the classifier model is a black box itself. how do we explain it suggests a text are 90% sure generated by AI?
 2. the model could be biased, e.g. against people use English as second language.
 3. easy to evade. by changing a few words, deliberately adding grammar errors, etc.

Watermarking provides identity of a texts by slightly biasing the distribution of each word during inference.



Why would this work?

Because no one knows the secret hash seed and function.

The tokens written by us (I mean anyone) can only have about 50% chance to be in red list or green list.

$P(\text{the } t\text{th word is in green list written by a human})$

$$= \sum_{\text{all words}} P(\text{t th word} = \text{word } x \text{ and word } x \text{ is in green list})$$

$$= \sum_{\text{all words}} \underline{P(\text{word } x \text{ is in green list})} P(\text{t th word} = \text{word } x)$$

$$= \frac{1}{2} \sum_{\text{all words}} P(\text{t th word} = \text{word } x) \quad \rightarrow \text{independent}$$

$$= \frac{1}{2}$$

The following is very standard hypothesis testing procedures.

Let S = tokens in green list

N = total tokens

then under the null hypothesis

H_0 : the texts are generated by human

$$Z = \frac{S - \frac{1}{2}N}{\sqrt{\frac{1}{4}N(1 - \frac{1}{2})}} \overset{\text{approximately}}{\sim} N(0, 1)$$

If, however the alternative hypothesis

H_1 : the texts are generated by OpenAI

$$Z \rightarrow +\infty$$

→ increase with sample size.

pros: guaranteed detection given
long sequence. very powerful

Cons: 1. Company exclusive

2. does not work well for short sequences.